

other costs like admin, marketing and distribution, it came to a total of 3.2 to 11.8 cents.

The above calculation led to a conclusion that wind power comes with a premium price. To overcome price limitations, the Canadian government provided some subsidies and incentives like one per cent of tax benefit per kilowatt hour (kWh) of electricity produced using wind power.

Thus it can be concluded that wind power still has a long way to go before mass adoption as the turbine cost has to come down. Not all governments can afford such a technology.

The case for India

India was among the first few countries in the world to establish the Ministry of New and Renewable Energy (MNRE). In the last two decades, the country has witnessed unprecedented growth in its wind power generation capacity from about less than 1GW (gigawatts) to more than 32.72GW till October 2017. India has the fourth largest installed wind power generation capacity in the world.

Of the total installed capacity for renewable energy generation, wind power accounts for about 10 per cent. It is seen that two-thirds of production happens between May to September months, coinciding with Monsoons. One of the major reasons behind the increase of wind power through the years has been the benefit of accelerated depreciation for businesses.

The levelised tariff for wind power

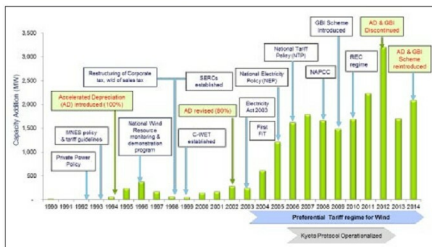


Fig. 1: Evolution of the Indian wind sector (Source: Ministry of New and Renewable Energy (2015b))

reached an all-time low of ₹ 2.64 per kWh during bidding for various wind power development projects in October 2017. Previously, the bid rate was ₹ 3.42 per kWh in August 2017 in an auction by TANGEDCO (Tamil Nadu Generation and Distribution Corporation).

The MNRE has set a target of producing about 60,000MW of wind power by the year 2022. To achieve this ambitious target, the government plans to enter into offshore wind power generation. It plans to set up the first plant along the Gujarat coast.

To conclude

At the end of 2006, worldwide capac-

Why do we need Wind Turbines?

- Wind power helps combat climate change and reduce reliance on fossil fuels.

Fun Fact!

Most wind energy comes from turbines that can be as tall as a 20-story building and have three 60-metre-long blades. Smaller turbines erected in a backyard can produce enough electricity for a single home or small business.

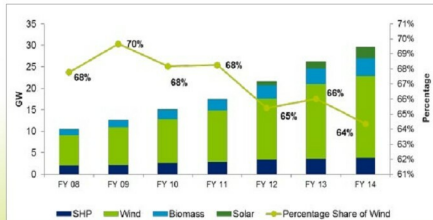


Fig. 2: Renewable energy capacity additions and share of wind (Source: Central Electricity Authority Report—Growth of Electricity Sector in India 1947-2013, FY 14 values as per MNRE)

ity of wind-powered generators was 73.9GW. Although it is believed that wind power presently produces a little above one per cent of world's electricity demand, it contributes around 20 per cent in Denmark, about 9 per cent in Spain and roughly 7 per cent in Germany. Around the world, wind power generation more than quadrupled between 2000 and 2006. There are many thousands of wind turbines operating, with a total capacity of 73,904MW, of which Europe accounts for 65 per cent (2006).

The most amazing part of wind power generation is no greenhouse gas emission. However, to derive the maximum benefit, wind power projects should be undertaken with long-term planning. To justify the heavy investment and variable output, these must be combined with other renewable technologies like reservoir-based hydro and gas plants. **EFY**